NEIGHBOURLY

Apartment Association Management System

Software Engineering Project | May 2026

Team-015 | BugsBunny

TEAM

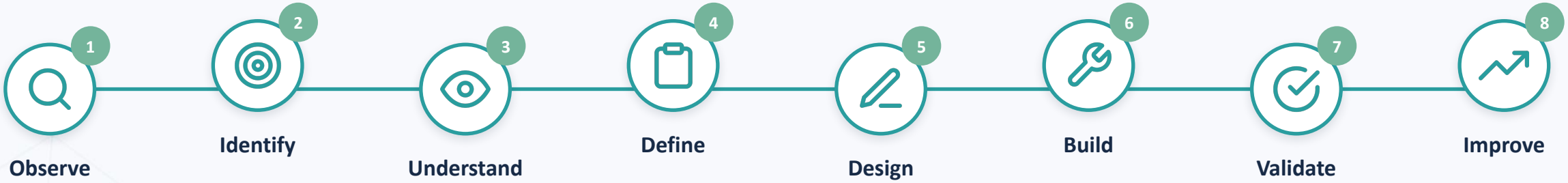
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From Everyday Friction to a Connected Community

How the project evolved

Neighbourly did not begin with a feature list. It began with a question: why does something as simple as apartment living involve so much friction? The answer emerged through a structured, human-centred process - one that shaped every decision that followed.

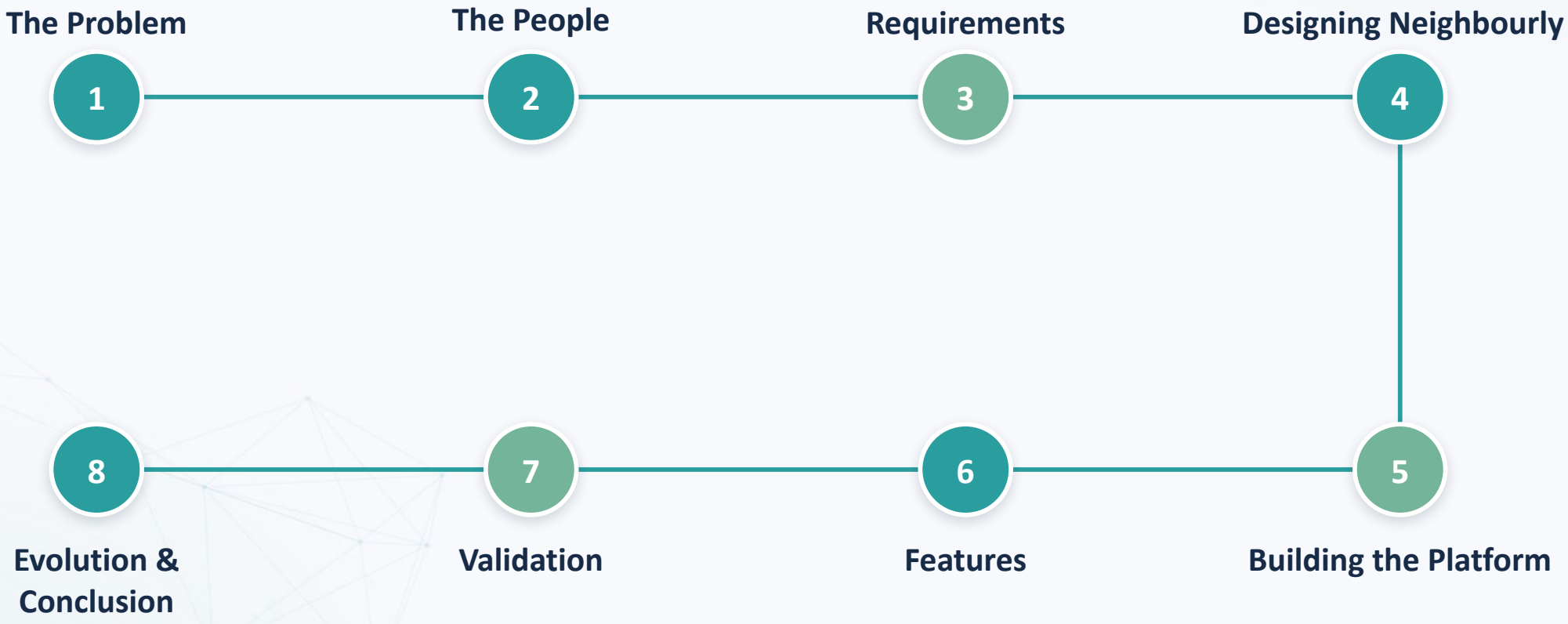


Every milestone of Neighbourly followed this arc - from listening closely to residents and administrators, to designing, building, and validating a platform that truly fits their needs.

Our Project Journey

The chapters that shape this presentation

Eight chapters carry the story of Neighbourly - from the problem we uncovered to the platform we built and validated.





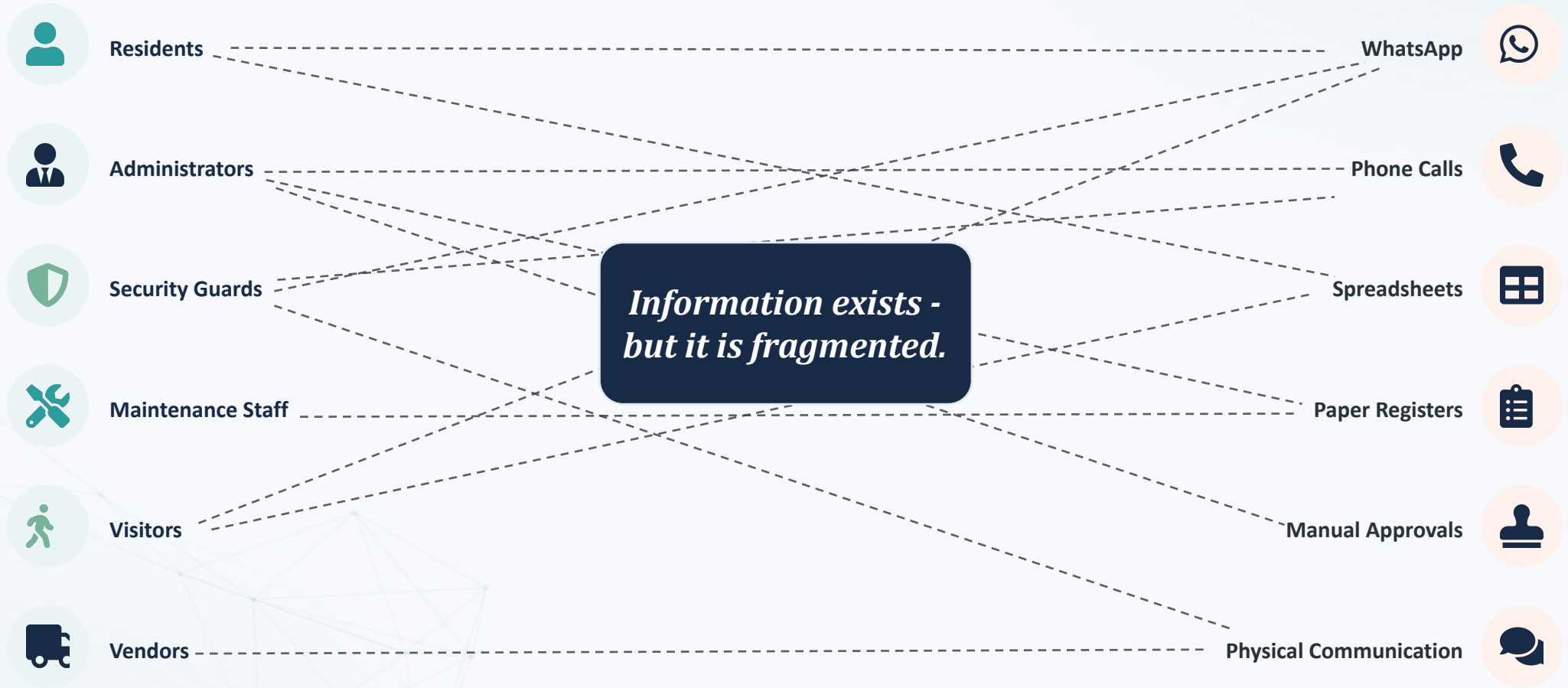
01

The Problem

Before building the solution, we first understood the everyday friction.

When Everything Works - Separately

Different people. Different channels. One community.



Where the Friction Happens

Eight recurring pain points surfaced across every stakeholder interview



Complaint Management

No structured way to report or track issues.



Communication

Key announcements get buried in group chats.



Visitor Management

Manual verification causes delays and queues.



Facility Booking

Availability checked only by contacting admins.



Record Management

Logs and history scattered across paper & sheets.



Transparency

Little visibility into progress or spending.



Administrative Workload

Admins coordinate everything by hand.



Community Participation

Voting & discussions leave no lasting record.

The Real Problem Wasn't the Lack of Technology

It was the lack of a single source of truth.

FRAGMENTED MANAGEMENT

- WhatsApp
- Phone Calls
- Spreadsheet
- Paper Register
- Manual Follow-up

WHAT THE COMMUNITY NEEDED

- One Platform
- Shared Information
- Trackable Workflows
- Role-Based Access
- Transparency

Fragmentation → Delays → Manual Effort → Poor Visibility



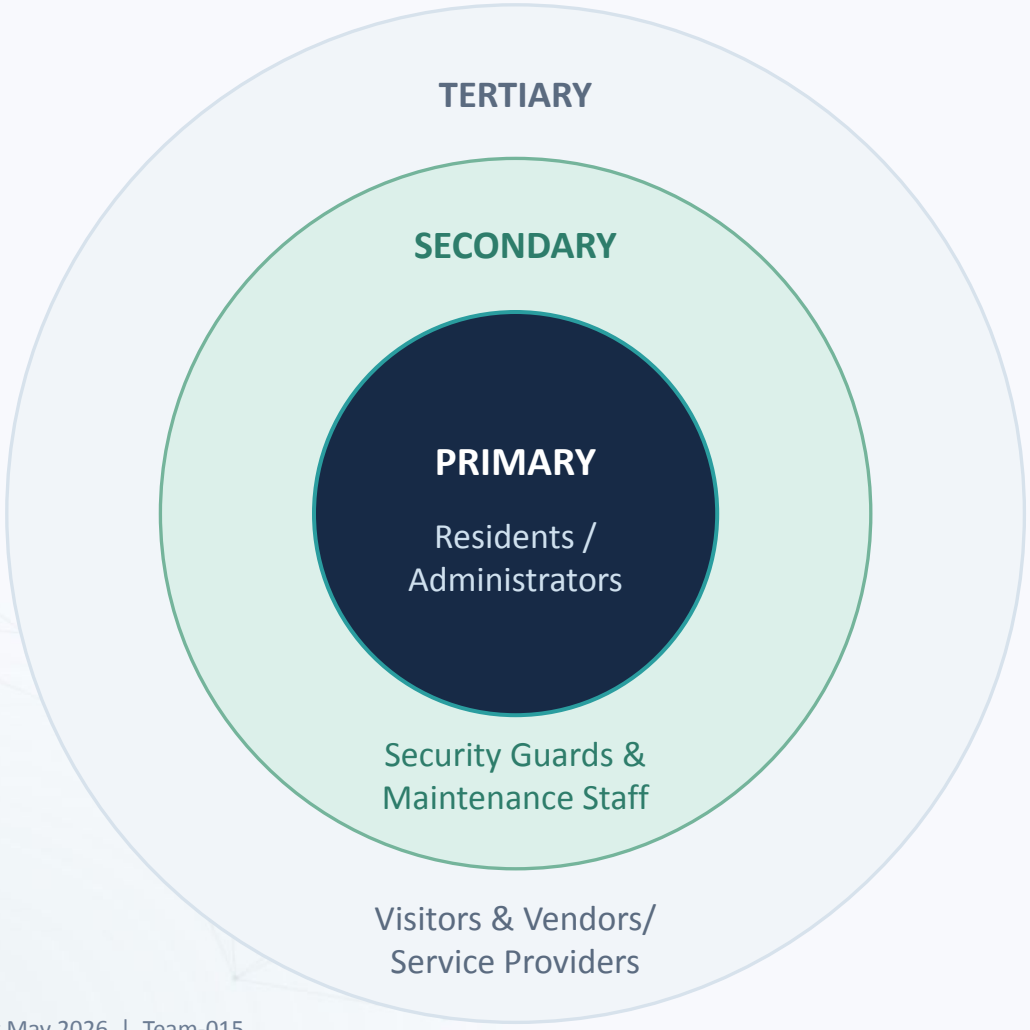
02

The People Behind the Requirements

A system is only useful when it fits the people who use it.

Who Uses the System?

A tiered stakeholder ecosystem, classified by frequency of interaction



PRIMARY USERS

- Residents / Apartment Owners
- Apartment Administrators / Association Committee

SECONDARY USERS

- Security Guards
- Maintenance Staff

TERTIARY USERS

- Visitors
- Vendors / Service Providers

Different Roles. Different Needs. One Community.

Every stakeholder interacts with Neighbourly differently - the platform adapts to each.



Residents

Complaints, facility booking, visitors, announcements.



Administrators

Operations, assignments, communication, management.



Security

Visitor verification and entry / exit records.



Maintenance

Assigned tasks and progress updates.



Visitors / Vendors

Verified and logged access.



03

Listening Before Building

We didn't start with features. We started with people.

We Went to the Users

Interviews and observations helped us understand how apartment operations actually work.

"It takes days to hear back..."



Semi-structured interviews · Workflow observation

Transcripts and recordings documented as supporting evidence.

- **Stakeholder Identification**
- **Interviews / Observation**
- **Pain Points**
- **Analysis**
- **Requirements**

Research conducted per Milestone 1: semi-structured interviews and workflow observation with residents and apartment administrators, supported by documented transcripts and recordings.

The Patterns Were Clear

Recurring themes across every interview and observation

01



Manual & Fragmented Processes

Workflows spread across WhatsApp, calls, spreadsheets, and paper registers — no centralized source of truth.

02



Lack of Transparency

Residents get little visibility into complaint progress, expenditure, or booking status.

03



Inefficient Communication

Announcements shared over WhatsApp are easily buried among regular conversations.

04



Heavy Manual Coordination

Complaint assignment, visitor approvals, and bookings depend heavily on available personnel.

05



Limited Digital Record-Keeping

Visitor logs, bookings, and maintenance records live in paper registers or spreadsheets.



The community needed a centralized platform connecting people, processes, and information.

Turning Pain Points Into a Problem We Could Solve

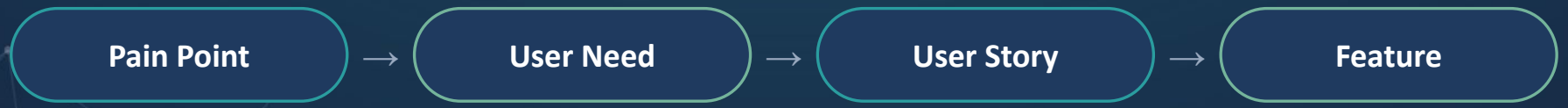
OBSERVED PAIN		UNDERLYING CAUSE		REQUIREMENT
Complaint follow-up by calls & chats	→	No structured workflow	→	Trackable complaint lifecycle
Announcements buried in chats	→	Fragmented communication	→	Centralized announcements
Visitor approval via phone calls	→	Manual verification	→	Digital visitor workflow
Booking needs admin coordination	→	No shared availability	→	Centralized facility booking
Records scattered across formats	→	Multiple storage locations	→	Centralized digital records

This became the foundation for Neighbourly.

04

From Problems to User Stories

Every requirement was expressed from the user's point of view.



Primary Users: What They Need

Representative stories from the Milestone 1 requirement baseline

Resident

As a resident, I want to view the status of my maintenance request, so that I can track its progress without repeatedly contacting the apartment office.

US-01

Resident

As a resident, I want to reserve shared facilities by selecting an available time slot, so that booking conflicts are avoided.

US-02

Resident

As a resident, I want to pre-register my visitors before their arrival, so that they can enter the apartment without unnecessary waiting.

US-03

Administrator

As an administrator, I want to assign maintenance requests to the appropriate staff, so that resident issues are resolved efficiently.

US-06

Administrator

As an administrator, I want to publish announcements from a centralized dashboard, so that all residents receive consistent information.

US-07

15 SMART user stories formed the initial requirement baseline.

Beyond Residents: The Whole Community Matters

Representative SMART user stories for secondary and tertiary stakeholders

SECONDARY USERS

Maintenance Staff — US-08



As a maintenance staff member, I want to view my assigned tasks and their priorities, so that I can complete them efficiently.

Security Guard — US-09



As a security guard, I want to verify visitor details against approved visitor records, so that only authorized visitors are allowed entry.

TERTIARY USERS

Visitor — US-11



As a visitor, I want my identity to be verified promptly, so that I can enter the apartment without unnecessary waiting.

Vendor — US-14



As a vendor, I want my service visit to be logged digitally, so that my access history is maintained accurately.

Requirements were defined across the entire apartment ecosystem - not just residents.

Generative AI User Stories

Features made with Generative AI and added subsequently

Administrator

As an administrator, I want AI to recommend complaint priority based on its description, so that urgent complaints can be addressed first.

US-16

Administrator

As an administrator, I want AI to recommend suitable staff while assigning a complaint, so that complaints are assigned to appropriate staff.

US-17

Admin/Committee Member

As an administrator/committee member, I want AI to summarize aggregated complaint and community data, so that I can quickly understand key trends and issues.

US-18

Resident

As a resident, I want to ask an AI chatbot about complaints and announcements, so that I can quickly find relevant community information.

US-19

Administrator

As an administrator, I want AI to suggest a category for a complaint based on its description, so that complaints can be categorized consistently.

US-20

5 SMART user stories were added during the later phases of the project



05

Meet Neighbourly

One platform for the everyday operations of a residential community.

Neighbourly at a Glance

Apartment Association Management System

Neighbourly is a web-based platform designed to digitize the day-to-day operations of a residential apartment community.



From Fragmented Workflows to One Connected System

The same operations, reimagined through a single connected platform

BEFORE

- Fragmented communication
- Manual complaint follow-up
- Manual visitor verification
- Scattered records
- Booking coordination
- Limited visibility
- Administrative effort

WITH NEIGHBOURLY

- **Centralized communication**
- **Trackable complaints**
- **Digital visitor workflows**
- **Centralized records**
- **Shared facility availability**
- **Role-specific visibility**
- **Streamlined operations**

06

Designing the Foundation

Before implementation, we designed how the system would work.

How Neighbourly Is Built

A layered architecture connecting client, API, and data

CLIENT LAYER

Frontend / User Interface

UI screens designed and prototyped in Figma across resident, admin, security and maintenance dashboards.

APPLICATION LAYER

Backend API - FastAPI

REST APIs built with FastAPI; JWT-based access & refresh tokens with Role-Based Access Control (RBAC) enforced via dependency injection.

FastAPI | JWT + RBAC | REST APIs

DATA LAYER

Database - PostgreSQL

Centralized, normalized relational database modeling 30 interconnected entities across all system modules.

API DOCUMENTATION, TESTING & DEVELOPMENT TOOLS

- Swagger / OpenAPI
- Pytest Testing
- Figma (Design)
- Jira (Sprints)
- GitHub (Version Control)

Designed to Be Modular

10 backend components, each owning a distinct functional area

- **Authentication & Authorization**
- **User & Apartment Management**
- **Complaint Management**
- **Maintenance Staff Management**
- **Visitor Management**
- **Facility Booking**
- **Communication & Notifications**
- **Billing & Payments**
- **Community Engagement**
- **AI Community Intelligence**

All 10 modules share a single PostgreSQL domain model - 30 interconnected entities linked through well-defined one-to-one, one-to-many, and many-to-one relationships (e.g., Resident → Complaints, Bill → Payments).



07

The Connected Community

The features are designed around the problems we discovered.

The Neighbourly Feature Ecosystem

The final feature set, grouped by purpose



From Complaint to Resolution

A trackable lifecycle from submission to closure



Resident submits complaint

Admin categorizes & assigns

Maintenance works on it

Status is updated

Resident tracks progress

Problem solved: Less follow-up. More visibility. Better accountability.

A Smoother Gate Experience

From manual verification to a digital, trackable visitor flow



How Neighbourly Helps

- Residents pre-register expected visitors with an OTP - guests no longer wait at the gate for a phone call.
- Security verifies visitor identity directly against the pre-registered request.
- Entry and exit timestamps are logged automatically for auditing and security records.

Faster, Verified Entry

Manual verification and waiting time at the gate are reduced through digital, pre-approved visitor records - with an optional workflow for delivery visitor tracking.

Book Community Spaces Without the Back-and-Forth

Real-time availability replaces phone-call coordination with the admin

View Availability

Real-time slot availability across facilities.

Select Slot

Pick an open time slot for the desired facility.

Reserve

System blocks the slot and prevents double-booking.

Confirmation

Resident receives booking confirmation.

Problem solved: reduces coordination and prevents conflicting bookings.

Clubhouse: Available Slots

9:00 – 10:00 AM	Available
10:00 – 11:00 AM	Booked
5:00 – 6:00 PM	Available
6:00 – 7:00 PM	Available

Gym | Swimming Pool | Turf | Clubhouse

From Group Chats to Structured Community Communication

Three connected capabilities replace scattered WhatsApp threads



Announcements

Centralized notices published by administrators from a single dashboard - replacing scattered WhatsApp messages.

Implemented in Milestone 4



Notifications

Real-time updates for complaint status, visitor approvals, facility bookings, and bill generation.

Unified notifications added in Milestone 4

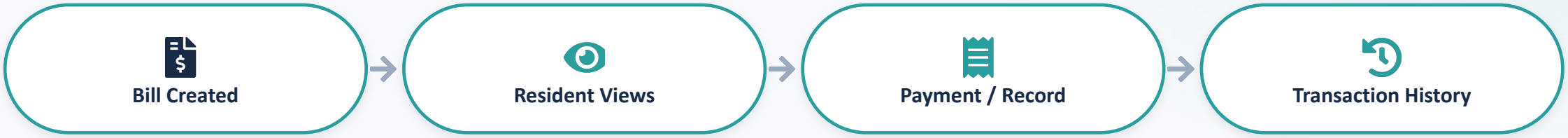


Community Engagement

Polls and voting let residents weigh in on community decisions, with live results after completion.

Community polling added in Milestone 4

Making Community Payments Easier



IMPLEMENTED

- Monthly maintenance bill generation for residents
- Outstanding dues & payment history visible to residents
- Residents record payments; admins view billing records
- Downloadable receipts and complete transaction history

FUTURE ENHANCEMENT

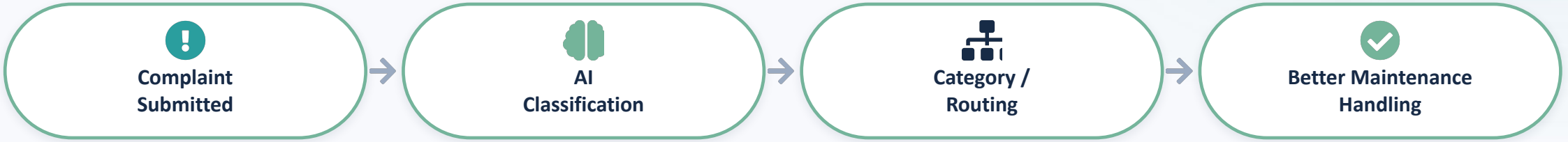
A real payment gateway integration remains pending, per Milestone 4 - online payments are currently recorded rather than processed through a live gateway.

SMS / email payment notifications are also pending for future sprints.

Making the Community Smarter

AI-assisted complaint handling is implemented and tested; broader intelligence is planned

IMPLEMENTED



FUTURE / PLANNED

Predictive Maintenance

Alerts based on complaint patterns.

Smart Staff Recommendations

AI-suggested assignee for each complaint.

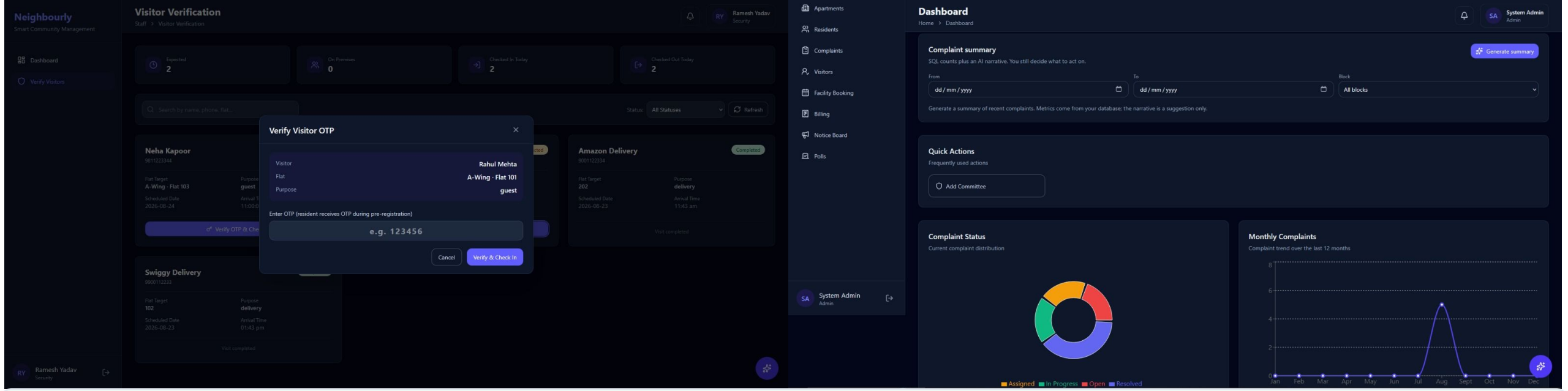
AI Visitor Verification

Facial recognition at the gate.

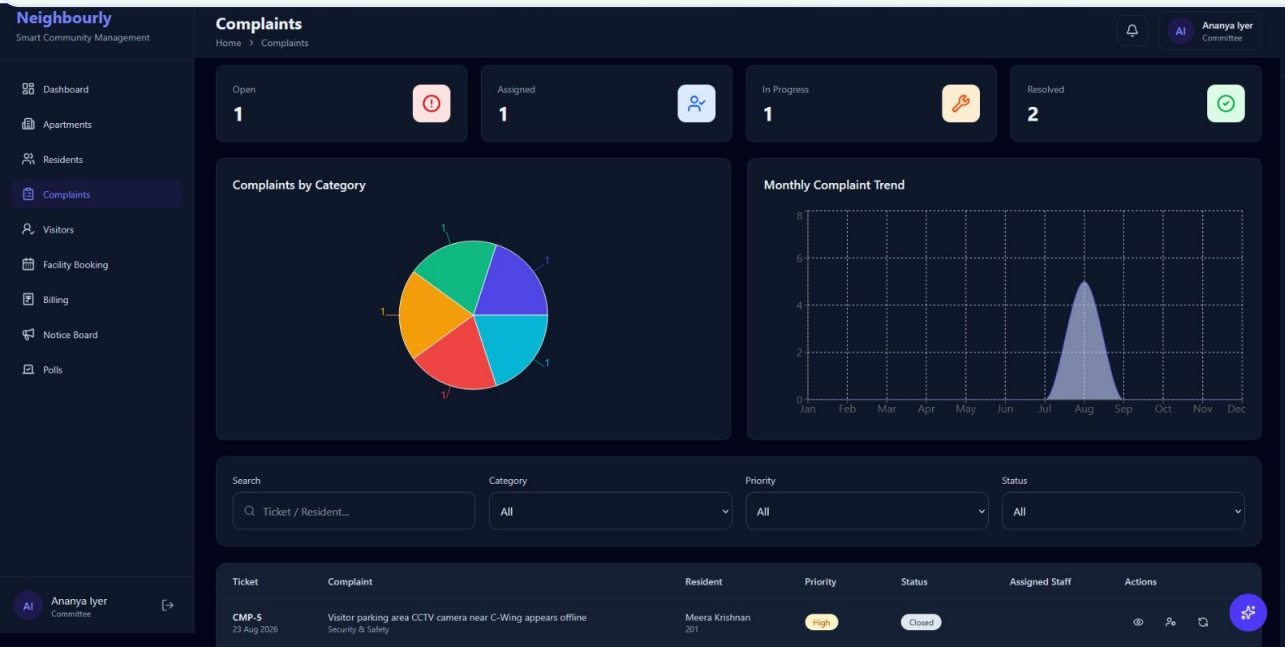
Energy Insights

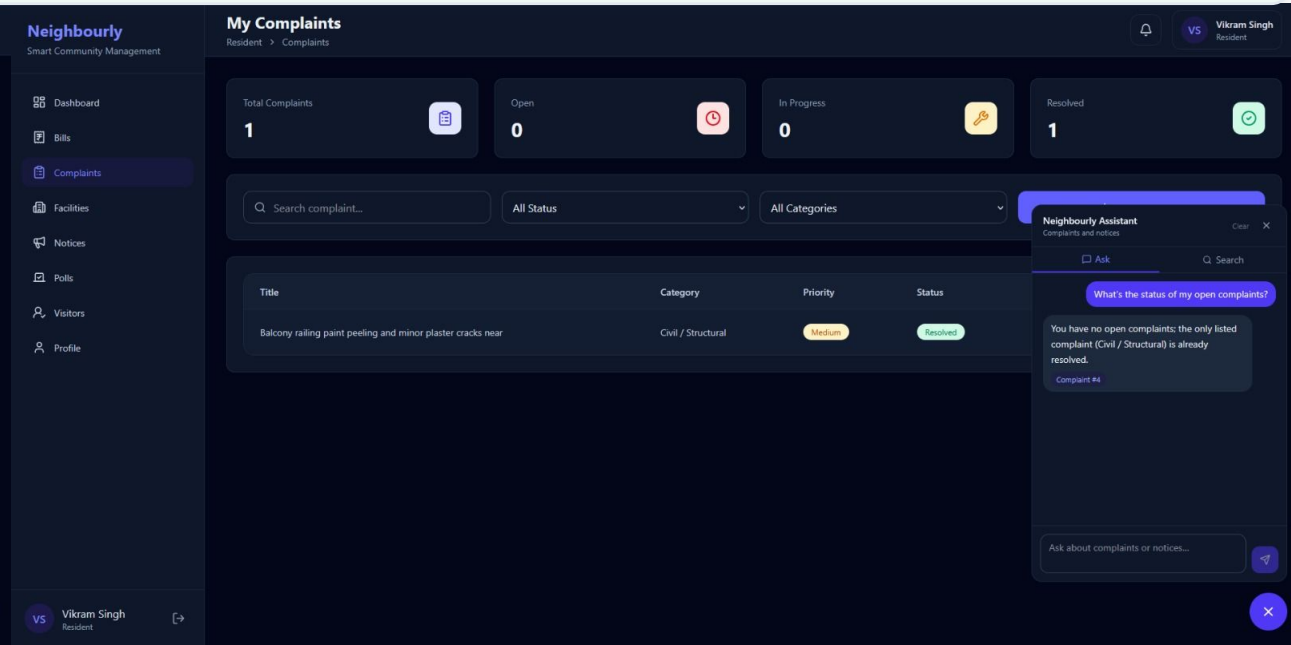
Smart energy consumption tracking.

AI-assisted complaint classification is live in Milestone 5. All other AI capabilities remain future enhancements, not yet implemented.



A few frontend screens (Resident getting OTP on registering visitors, Admin Dashboard, Committee Member Dashboard, AI Chatbot

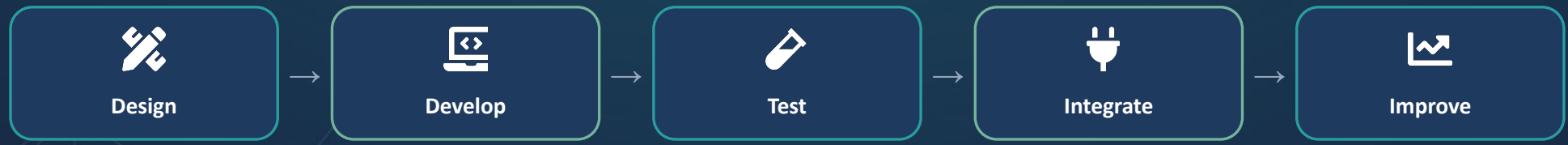




08

From Design to Working System

The project moved from blueprints to APIs, integration and validation.



How the Project Progressed

Agile Scrum development, organized into five milestone-based deliverables

User Requirements

15 Jun – 28 Jun 2026

- Stakeholder interviews
- Pain point analysis
- SMART user stories
- Product backlog

M1

M2

Planning & System Design

29 Jun – 22 Jul 2026

- Component & class design
- ER diagram
- Sprint schedule & Kanban
- Frontend prototype

M3

Backend Development

23 Jul – 02 Aug 2026

- Auth, Complaint, Visitor APIs
- Booking & Billing APIs
- Swagger documentation
- Backend testing

M4

Integration & System Testing

03 Aug – 12 Aug 2026

- Frontend–backend integration
- System testing & bug fixes
- UI refinements
- Sprint review & retrospective

M5

Final Submission

13 Aug – 23 Aug 2026

- Final documentation & README
- Code packaging
- Presentation & demo video
- Final submission

From “It Works” to “We Tested It”

Milestone 4 validation results - hybrid pytest and Swagger-driven contract testing

97	10	103	0	0
OpenAPI Operations	Backend Modules	Passing Test Cases	Failing Test Cases	Unverified Operations

VALIDATION APPROACH

pytest + httpx
API testing — 103 async test cases with SQLite in-memory DB

Swagger / OpenAPI
API contract validation across all 97 endpoints (OpenAPI 3.1.0)

Pydantic / RBAC
Schema validation & role-based security enforcement

Error-Path Testing
401 / 403 / 404 / 409 / 422 / 503 scenarios verified

Milestone 4 expanded the API surface by 160% (30 → 97 endpoints, 7 → 10 modules) from Milestone 3, with full pytest evidence for every endpoint.

Feedback Changed the Product

What residents and administrators told us in Milestone 3, and what we built in response

USER FEEDBACK / NEEDS	WHAT WE CHANGED
Better complaint-update notifications	**Unified notifications system**
Centralized announcements	**Centralized announcement dashboard**
Maintenance bill handling	**Billing & payment functionality**
Improved visitor approval	**Visitor workflow improvements**
Community polls / voting	**Community polling feature**
	AI-assisted complaint classification
	Maintenance workflow improvements

Testing did not end with validation — feedback became input for the next iteration.



09

What We Built & What We Learned

Neighbourly: Connecting the Community, Not Just the Software

01

User-Centered

Started with stakeholders, interviews, and real pain points.

02

Structured

Converted findings into SMART user stories and requirements.

03

Modular

Designed a scalable system around core apartment workflows.

04

Validated

Implemented and tested a growing backend with strong API/test coverage.

97	103	0
API Operations	Passing Tests	Failures

Neighbourly transforms fragmented apartment operations into a connected, transparent and more manageable digital ecosystem.

Where Neighbourly Goes Next

FUTURE ENHANCEMENTS

Not yet implemented - a forward-looking roadmap

Real Payment Gateway Integration

Live payment processing beyond manual recording.

SMS / Email Notification Integration

Delivering alerts beyond in-app notifications.

Expanded AI Community Intelligence

Smart search and an AI chat assistant for residents.

Predictive Maintenance

Alerts generated from historical complaint patterns.

Smart Visitor Verification

Facial recognition support at the gate.

Energy Insights

Smart energy consumption tracking and recommendations.

Additional Community Engagement

Deeper tools for resident participation.



Thank You